

 **TARGET CAT 2025** 



MBA RAPID FIRE

QUANT : Algebra

Algebra
Part I

Lecture No.- 01

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TOPICS *to be covered*



- 1) Revision & Numerical based on 'Exponents'
- 2) Revision & Numerical based on 'Linear Equations'
- 3) Revision & Numerical based on 'Quadratic Equations'



Algebraic Identities :

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $a^2 - b^2 = (a + b)(a - b)$
- $a^2 + b^2 = (a + b)^2 - 2ab$ or $a^2 + b^2 = (a - b)^2 + 2ab$
- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
- $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$



Exponents



Properties of Exponents :



Properties of Exponents :

1. $a^m \cdot a^n = a^{(m+n)}$

2. $(a^m)^n = a^{m \cdot n}$

3. $a^m / a^n = a^{(m-n)}$

4. $\sqrt{(a^m)} = a^{(m/2)}$, $\sqrt[3]{(a^m)} = a^{(m/3)}$, $\sqrt[4]{(a^m)} = a^{(m/4)}$

5. $\boxed{a^m = 1/(a^{-m})}$ & $\underline{a}^{-m} = 1/(\underline{a}^m)$

6. $\underline{(x \cdot y)^n} = x^n \cdot y^n$ & $\left(\frac{x}{y}\right)^n = \frac{x^n}{y^n}$

$$(-1)^{(-4)} = \frac{1}{(-1)^4} = \frac{1}{1} = 1$$

7. $(\pm a)^0 = 1$, $(1)^{\pm a} = 1$, $(-1)^{even} = 1$, $(-1)^{odd} = -1$

8. If $a^m = b$ then $a = b^{1/m}$

#Q. If $x = 1 + \frac{1}{\sqrt{2} + \sqrt{1}} + \frac{1}{\sqrt{3} + \sqrt{2}} + \frac{1}{\sqrt{4} + \sqrt{3}} + \dots + \frac{1}{\sqrt{1024} + \sqrt{1023}}$.

Also $\sqrt{x\sqrt{x\sqrt{x\ldots\infty}}}=a^b$, where a and b are integers, then find the value of (a + b). Minimum
↑
2 + 5 = 7

$$x = 1 + \sqrt{2} - 1 + \sqrt{3} - \sqrt{2} + \sqrt{4} - \sqrt{3} + \dots + \sqrt{1024} - \sqrt{1023}$$

$$x = \sqrt{1024} = \underline{\underline{32}}$$

$$\boxed{\frac{1}{\sqrt{2}+1}} = \frac{(\sqrt{2}-1)}{(\sqrt{2}+1)(\sqrt{2}-1)} = \boxed{\sqrt{2}-1}$$

$$\frac{1}{\sqrt{n+1} + \sqrt{n}} = \sqrt{n+1} - \sqrt{n}$$

$$\sqrt{x \sqrt{x \sqrt{x \dots \infty}}} = y = 32$$

$$x \sqrt{x \sqrt{x \sqrt{x \dots \infty}}} = y^2$$

$$\underline{xy = yx}$$

$$x = y = 32$$

$$\sqrt{x \sqrt{x \sqrt{x \dots \infty}}} = 32$$

$$a^b = 2^5$$

$$a^b = 32'$$

$$\begin{array}{r} a = 2 \\ b = 5 \\ \hline 7 \end{array}$$

QUESTION- 2

$$\sqrt{\quad} = (\quad)^{\frac{1}{2}}$$

$$(a^m)^n = (a^n)^m = a^{mn}$$



$$9^{m-n} = \frac{9^m}{9^n}$$

#Q. If $2^{6x} = 12100$, $3^{\sqrt{y}} = \sqrt{11}$ and $(\sqrt{7})^z = \sqrt{\frac{5}{8}}$ then evaluate the value of

$$\frac{2^{3(x-1)} \times 9^{-\sqrt{y}}}{49^z}$$

$$\frac{55}{4} \times \frac{1}{4} \times \frac{8}{8} \text{ (2)}$$

$$2^{6x} = 12100$$

$$(2^{6x})^{\frac{1}{2}} = (121 \times 100)^{\frac{1}{2}}$$

$$\frac{2^{3x}}{2} = 11 \times 10 = 110$$

$$2^{3(x-1)} = 2^{3x-3}$$

$$= \frac{2^{3x}}{2^3}$$

$$= \frac{110}{8} = \frac{55}{4}$$

$$3^{\sqrt{y}} = \sqrt{11}$$

$$(3^{\sqrt{y}})^2 = (\sqrt{11})^2$$

$$(3^2)^{\sqrt{y}} = 11$$

$$9^{\sqrt{y}} = 11$$

$$9^{-\sqrt{y}} = \frac{1}{9^{\sqrt{y}}}$$

$$9^{-\sqrt{y}} = \frac{1}{11}$$

$$(7^{\frac{1}{2}})^z = \sqrt{\left(\frac{5}{8}\right)^{\frac{1}{2}}}$$

$$(7^{\frac{1}{2}z}) = \left(\frac{5}{8}\right)^{\frac{1}{4}}$$

$$\left(7^{\frac{1}{2}z}\right)^{\frac{1}{2}} = \left[\left(\frac{5}{8}\right)^{\frac{1}{4}}\right]^{\frac{1}{2}}$$

$$(7^z)^{\frac{1}{2}} = \frac{5}{8}$$

$$(7^z)^{\frac{1}{2}} = \frac{5}{8}$$

A 4

B 1

C 2

D 3



Linear Equation



Linear Equation :

$$ax + b = 0$$

$$ax + by = c$$

$$ax + by + cz = d$$

① Method of Elimination

② Method of substitution

$$x + y = 8$$

$$2x - y = 6$$



Types of Solution in Linear Equation :

If there are two linear equations,

_____ L
_____ M

$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned}$$

(overlapping lines)

Unique Solution

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$$



Infinite Solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$



Solⁿ

$$\begin{aligned} x &= ? \\ y &= ? \end{aligned}$$

(parallel lines)

No Solution

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

QUESTION-3

$$a, b, c \geq 1$$

$$c = 1 \rightarrow \text{min}$$



#Q. Sarita's mother gave Sarita Rs. 50, Rs. 200 and Rs. 500 notes to buy some groceries from the market. There is at least one note of every denomination. Sarita has exactly 200 notes with total amount as Rs. 12550. Then, the maximum number of Rs. 200 notes Sarita was given is,

	<u>No. of Notes</u>
Rs. 50 $\rightarrow$	a
Rs. 200 $\rightarrow$	b ? (max)
Rs. 500 $\rightarrow$	c
	<u>200</u>

$$a + b + c = 200$$

$$50a + 200b + 500c = 12550$$

$$a + 4b + 10c = 251$$

$$3b + 9c = 51$$

$$b + 3c = 17$$

$$b = 17 - 3c$$

$$b = 17 - 3(1)$$

$$b = 14$$

$$17 - 3 = 14$$

$$c \downarrow b \uparrow$$

QUESTION- 4



#Q. A laboratory uses 5 beakers of solution A, 6 beakers of solution B and 6 beakers of solution C, which costs the same as using 4 beakers of solution A, 6 beakers of solution B and 9 beakers of solution C or using 9 beakers of solution B and 8 beakers of solution C. The cost of how many beakers of solution C is the same as the cost of any one of the set of entire solutions?

A 47

B 48

C 49

D 50

let, Solⁿ A $\rightarrow$ a
 Solⁿ B $\rightarrow$ b
 Solⁿ C $\rightarrow$ c

CP/beaker

$$5a + 6b + 6c = 4a + 6b + 9c = 9b + 8c$$

$$5a + 6b + 6c = 4a + 6b + 9c$$

$$a = 3c$$

$$5(3c) + 6\left(\frac{13}{3}c\right) + 6c = 15c + 26c + 6c = 47c$$

$$4a + 6b + 9c = 9b + 8c$$

$$12c + 6b + 9c = 9b + 8c$$

$$21c + 6b = 9b + 8c$$

$$13c = 3b$$

$$\frac{13}{3}c = b$$

QUESTION- 5

T. Biscuits = x



#Q. A lady decides to distribute a pile of biscuits among a few street dogs in the following manner :

She gives one-third of her current stock plus 16 extra biscuits to the first dog. To the second dog, she gives one-third of the remaining stock plus 16 extra. She continues this process for a total of 4 dogs, giving each one-third of the remaining stock plus 16 extra biscuits every time. After feeding the fourth dog, she is left with no biscuits.

What was the initial number of biscuits she had ?

$$16 \times \frac{3}{2}$$

$$40 \times \frac{3}{2}$$

$$76 \times \frac{3}{2}$$

$$\frac{2}{3} \left\{ \frac{2}{3} \left[\frac{2}{3} \left(\frac{2}{3}x - 16 \right) - 16 \right] - 16 \right\} - 16 = 0$$

$$\frac{2}{3} \left[\frac{2}{3} \left(\frac{2}{3}x - 16 \right) - 16 \right] - 16 = 25$$

$$\frac{2}{3} \left(\frac{2}{3}x - 16 \right) - 16 = 60$$

$$\frac{2}{3}x - 16 = 115$$

$$\frac{2}{3}x = 130$$

$$x = \frac{130 \times 3}{2}$$

$$x = 195$$

QUESTION- 6



#Q. For some real numbers p and q , the system of equations $4a + 9b = 15$ and $(p + 9)a + (q^2 + 80)b = 30q$ has infinitely many solutions for a and b . Then the maximum possible value of $p - q + pq$

771

$$\underline{4a + 9b = 15} \quad | \quad \underline{(p+9)a + (q^2+80)b = 30q}$$

$$\frac{p+9}{4} = \frac{\frac{q^2+80}{9} = \frac{30q}{15}}$$

$$\frac{q^2+80}{9} = 2q$$

$$q^2 - 18q + 80 = 0$$

$$\begin{aligned} x &\rightarrow a=1, b=-18, c=80 \\ + &\rightarrow -18 \end{aligned}$$

$$\frac{p+9}{4} = 2q$$

$$p+9 = 8q$$

$$p = 8q - 9$$

$$q = 10$$

$$\text{or } q = 8$$

$$q = 10$$

$$p = 8(10) - 9$$

$$= 71$$

$$p - q + pq$$

$$= 71 - 10 + 710$$

$$= 771$$

$$q = 8$$

$$p = 8(8) - 9$$

$$p = 55$$

$$p - q + pq$$

$$55 - 8 + 440$$

$$= 487$$



Quadratic Equation



Quadratic Equation :



$$a \neq 0$$

$$ax^2 + bx + c = 0$$

Solve $\rightarrow$ x values $\rightarrow 2$

Roots of Eqn

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$3x^2 + 8x + 4 = 0$$

$$a = 3, b = 8, c = 4$$

$$x \rightarrow (a \times c) = 12$$

$$+ \rightarrow b = 8 \begin{matrix} 6 \\ 2 \end{matrix}$$

$$x = \frac{6}{-3}$$

$$= \frac{-6}{3}$$

$$= -2$$

$$x = \frac{2}{-3}$$

$$= \frac{-2}{3}$$

$$= -\frac{2}{3}$$



Quadratic Equation :



Roots
 α & β

$$k \cdot [x^2 - (\alpha + \beta)x + \alpha\beta] = 0$$

5 & -2

$$x^2 - 3x + (-10) = 0$$

$$\begin{array}{l} (k=2) \times \left(\begin{array}{l} x^2 - 3x - 10 = 0 \\ 2x^2 - 6x - 20 = 0 \end{array} \right) \end{array}$$

$\times (k=-3)$

$$-3x^2 + 9x + 30 = 0$$

$$ax^2 + bx + c = 0$$

$$\alpha + \beta = \frac{-b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Discriminant = $b^2 - 4ac$
(Nature of Roots)

Quadratic Equation :



Points to Remember :

1. Quadratic Equations are represented as

$$(ax^2 + bx + c = 0) \text{ where } a \neq 0$$

Roots are Real

$$D \geq 0$$

2. Discriminant (D) = $b^2 - 4ac$ (Distinct)

* If $D > 0$ ----- Roots are Real & Unequal

* If $D = 0$ ----- Roots are Real & Equal

* If $D < 0$ ----- Roots are Imaginary (Complex Roots or No Real Roots)

3. If one of the roots of Quadratic Equation is $(x + \sqrt{y})$, then other root can be represented as $(x - \sqrt{y})$ & vice versa.

4. If α & β are the roots of quadratic equation $ax^2 + bx + c = 0$, then

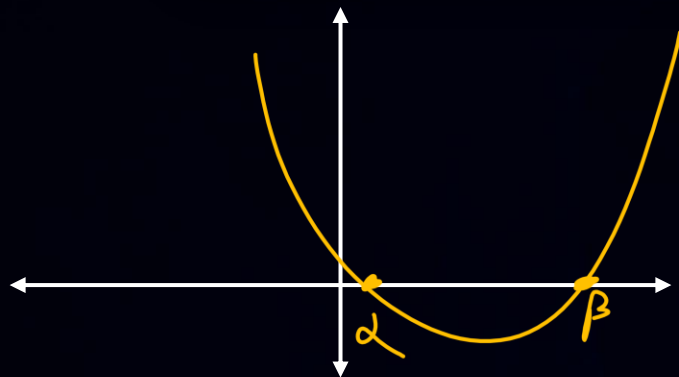
$$\text{Sum of Roots } (\alpha + \beta) = \frac{-b}{a} \quad \& \quad \text{Product of Roots } (\alpha\beta) = \frac{c}{a}$$

5. If α & β are the roots of quadratic equation, then equation can be represented as

$$\underline{k[x^2 - (\alpha + \beta)x + \alpha\beta] = 0}$$

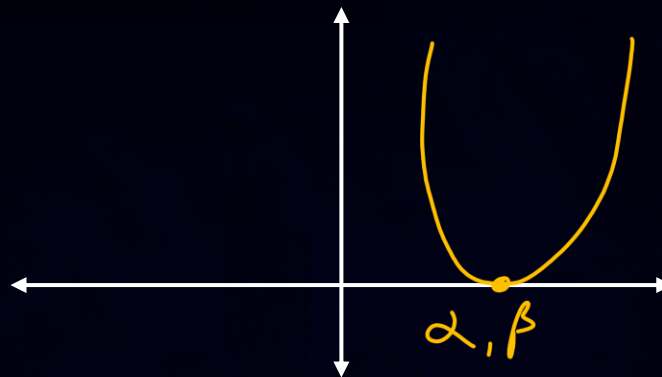
$$\underline{ax^2 + bx + c}$$

Graphical Approach :



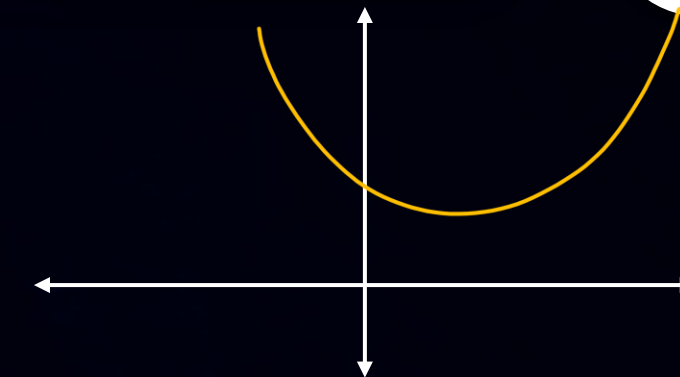
$$\underline{a > 0}; D > 0$$

Real and Unequal Roots



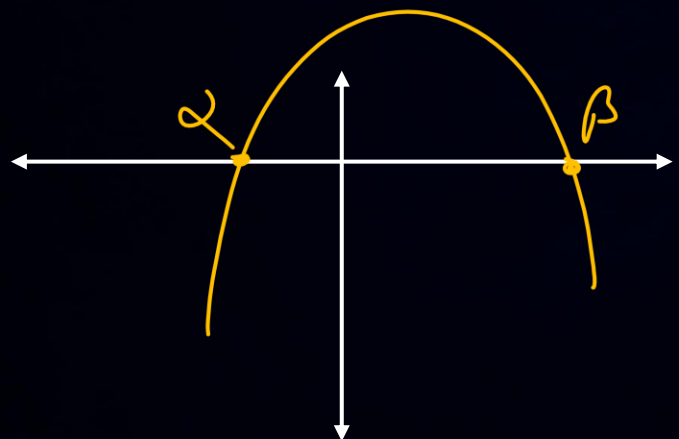
$$\underline{a > 0}; \underline{D = 0}$$

Real and equal Roots

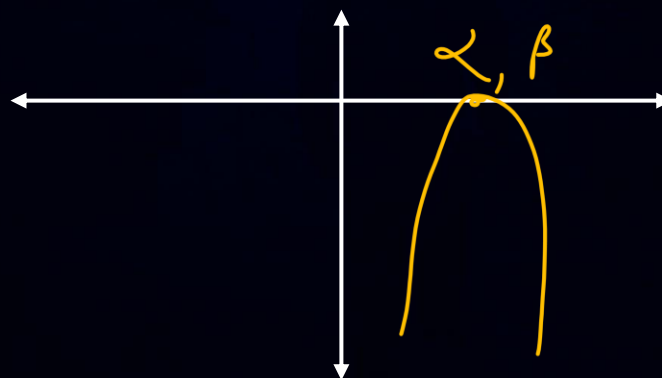


$$\underline{a > 0}; D < 0$$

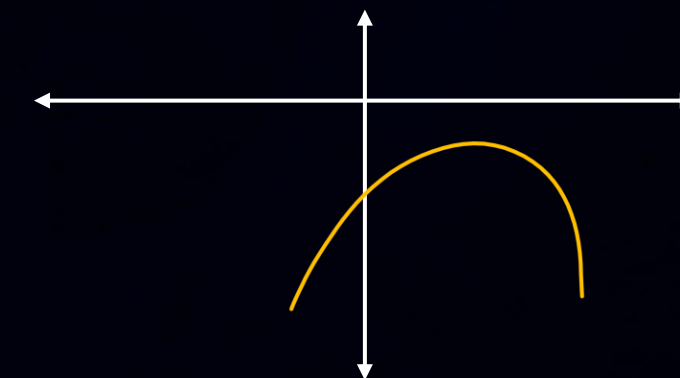
No Real Roots



$$\underline{a < 0}; \underline{D > 0}$$



$$\underline{a < 0}; D = 0$$



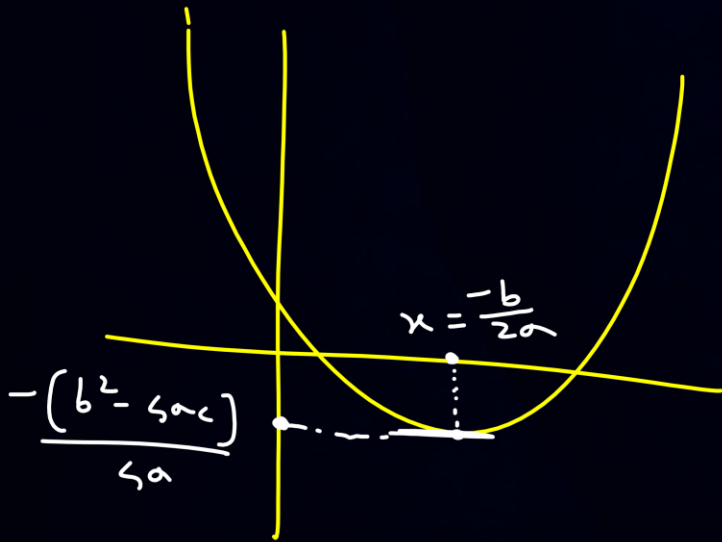
$$a < 0; D < 0$$



Maximum & Minimum Value of QE:



$$a > 0 \quad y = \underline{ax^2 + bx + c}$$

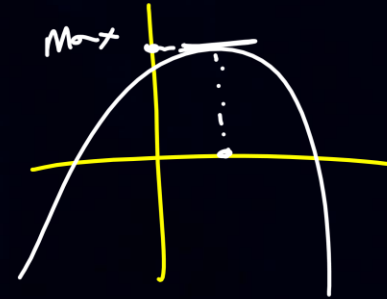


$$\text{Max} \rightarrow \infty$$

$$\text{Min} = \underline{-\frac{[b^2 - 4ac]}{4a}}$$

$$\text{at } x = -\frac{b}{2a}$$

$$\underline{a < 0}$$



$$\text{Min} \rightarrow -\infty$$

$$\text{Max} \rightarrow \underline{-\frac{[b^2 - 4ac]}{4a}}$$

$$\text{at } x = -\frac{b}{2a}$$

QUESTION- 7



#Q. If -5 is a root of the quadratic equation $2x^2 + px - 15 = 0$ and the quadratic equation $p(x^2 + x) + k = 0$ has equal roots, find the value of k .

$$2x^2 + px - 15 = 0$$

$$p \text{ and } x = -5$$

$$\rightarrow 2(-5)^2 + p(-5) - 15 = 0$$

$$-5p + 35 = 0$$

$$\cancel{35} = \cancel{5}p$$

$$7 = p$$

$$p(\underline{x^2 + x}) + k = 0$$

$$7x^2 + 7x + k = 0$$

$$a = 7, b = 7, c = k$$

$$b^2 - 4ac = 0$$

$$7^2 - 4(7)(k) = 0$$

$$49 = 28k$$

$$\frac{\cancel{49}}{\cancel{28}} = k$$

$$\frac{7}{4} = k$$

A $-49/12$

B $7/4$

C $-7/4$

D $49/12$

QUESTION- 8

$$a^m \cdot a^n = a^{m+n}$$



#Q. Let α and α^2 be the roots of $x^2 + x + 1 = 0$, then find the value of $\alpha^{31} + \alpha^{62} + \alpha^{93}$

$$a=1, b=1, c=1$$

$$\alpha^{62} = \alpha^{31} \cdot \alpha^{31}$$

$$\alpha + \alpha^2 = -\frac{b}{a} = -\frac{1}{1} = -1$$

$$\alpha \cdot \alpha^2 = \frac{c}{a} = \frac{1}{1} = 1$$

$$\alpha^3 = 1$$

A 1

B -1

C 0

D -1/2

$$\alpha^{31} + \alpha^{62} + \alpha^{93}$$

$$= \alpha^{31} [1 + \alpha^{31}] + (\alpha^3)^{31}$$

$$= \alpha^1 \cdot \alpha^{30} [1 + \alpha^1 \cdot \alpha^{30}] + (1)^{31}$$

$$= \alpha \cdot (\alpha^3)^{10} [1 + \alpha \cdot (\alpha^3)^{10}] + 1$$

$$= \alpha [1 + \alpha] + 1$$

$$= (\alpha + \alpha^2) + 1$$

$$= -1 + 1 = 0$$

QUESTION- 9



$$ax^2 + bx + c = 0 \quad a=1 \quad b=(3-p) \quad c=-p$$

#Q. Let α and β be the roots of $x^2 + (3-p)x - p = 0$. The value of p for which $\alpha^2 + \beta^2$ is minimum, is

$$\alpha + \beta = -\frac{b}{a} = -\frac{(3-p)}{1} = p-3$$

$$\alpha\beta = \frac{c}{a} = \frac{-p}{1} = -p$$

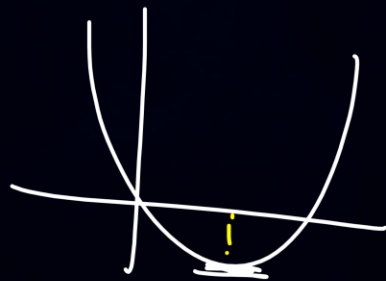
$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

$$= (p-3)^2 - 2(-p)$$

$$= p^2 - 6p + 9 + 2p$$

$$\alpha^2 + \beta^2 = p^2 - 4p + 9 \quad a=1 \quad b=-4$$

$$a > 0$$



$$\begin{aligned} \text{at } p &= \frac{-b}{2a} \\ &= \frac{-(-4)}{2(1)} = \frac{4}{2} = 2 \end{aligned}$$

A 0

B 1

C 2

D 3

QUESTION- 10



#Q. Consider the quadratic equation $3x^2 + 3qx + p = 0$, where p and q are positive integers and the equation has real roots. If q is in the set $\{2, 3\}$ and p is in the set $\{1, 2, 3, 4\}$, how many pairs of (p, q) are possible?

$$3x^2 + 3qx + p = 0$$

$$a = 3, b = 3q, c = p$$

$$D \geq 0$$

$$b^2 - 4ac \geq 0$$

$$(3q)^2 - 4(3)(p) \geq 0$$

$$9q^2 - 12p \geq 0$$

$$9q^2 \geq 12p$$

$$9q^2 \geq 12p$$

$$put \ q = 2$$

$$9(4) \geq 12p$$

$$3 \geq p$$

$$(1, 2), (2, 2), (3, 2)$$

$$put \ q = 3$$

$$9(9) \geq 12p$$

$$6.75 \geq p$$

$$(1, 3), (2, 3), (3, 3), (4, 3)$$

A 5

B 6

C 7

D 8

QUESTION- 11



#Q. If both the roots of the quadratic equation $x^2 - 375x + k = 0$ are prime numbers, the sum of all possible values of 'k' is

746

Roots $\rightarrow \alpha$ & β

2 & 373

$$a = 1, b = -375, c = k$$

$$\alpha + \beta = -\frac{b}{a} = 375$$

$$2 + \beta = 375 \Rightarrow \beta = 373$$

$$\alpha\beta = k$$

$$2 + 373 = k$$

$$746 = k$$

2, 3, 5, 7, 11, 13, 17, 19, ...

7 + 11 $\rightarrow$ even

13 + 19 $\rightarrow$ even

$$\text{Even} + \text{odd} = \text{odd}$$

$$2 + \text{odd} = 375$$

$$\text{odd} = 373$$

QUESTION- 12



#Q. If the minimum value of the quadratic expression $x^2 + ax + 12 = 0$ is equal to the minimum value of the quadratic expression $x^2 + bx - 3 = 0$, how many integer pairs (a, b) satisfy this condition ?

TRY
(Next Class)

QUESTION- 13



#Q. If x is real, then the sum of maximum and minimum values of the expression

$\frac{x^2-3x+4}{x^2+3x+4}$ will be

A $50/7$

B $49/7$

C $7/49$

D $7/50$



2 Mins Summary

- 1) Revision & Numerical based on 'Exponents'
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- 3) Revision & Numerical based on 'Quadratic Equations'



*Thank
You*